

Courses #1	Period	Pre-requisites	Co-requisites	Special requirements
BIO2001 Cell Biology	1	none	none	
BIO2004 General Zoology	1	none	none	
BIO2007 Genetics	1	BIO2001	PRA2024	
CHE1101 Introduction to Chemistry	1	none	PRA1101	
CHE2004 Spectroscopy	1	CHE2001 or CHE2009	none	
CHE2009 Fundamentals of Organic Chemistry	1	none	PRA2032	NOTE: This course cannot be followed in case you already followed CHE2001 Organic Chemistry
CHE3006 Quantum Chemistry	1	MAT2006 or MAT2009	PRA3018	MAT2004 and/or MAT2008 and/or MAT2009
CHE3010 Inorganic Chemistry	1	CHE2010	PRA3028	The course integrates nicely with Spectroscopy CHE2004. NOTE: Waivers can be granted knowing that there is material you will not be familiar with if you have passed CHE2009 Fundamentals of Organic Chemistry.
INT2013 Fundamentals of Science Education	1	PRO1002: Project Period (1000-level)	none	
INT3003 The Philosophy of Technology	1	PHI1002	none	
INT3002 Advanced Microscopy: Theory and Applications	1	none	none	PHY1001 or PHY1105
INT3014 Conservation Paleobiology	1	PRA1008 or INT1007	none	
INT3201 Island Studies: Life and Landscape	1	PRA2009 + INT1007 + INT2012 + Previously passed + 3000 module in either Biology, Archeology, or Paleontology.	INT2022 + PRA3201	
INT3202 Island Studies: The Mediterranean Region	1	PRA2009 + INT1007 + INT2012 + Previously passed + 3000 module in either Biology, Archeology, or Paleontology.	INT3201 + PRA3201	
MAT2007 Introduction to Programming	1	none	none	
NEU2003 Neuroethology	1	NEU1001 or NEU1002 or BIO3004	none	
NEU2005 Systems Neuroscience	1	none	none	Recommended: NEU1001 + Basic understanding of genetics, e.g. high school biology or BIO2007
PHY1101 Introduction to Physics	1	none	none	
PHY2004 Electronics	1	none	PRA2006	
PHY2009 Stellar Astronomy	1	none	none	
PHY2012 Structure of Matter	1	PHY2002	none	
PHY3001 Quantum Mechanics	1	PHY2005	none	Recommended: MAT2006 / MAT2004
PHY3009 Hamiltonian & Lagrangian Mechanics	1	PHY2001 or PHY1003	none	Recommended: PHY2011 + A course containing quantum theory
PRA1101 Introduction to Scientific Research I	1	MSF1000	CHE1101 + PHY1101	
PRA2003 Programming	1	none	MAT2007	
PRA2006 Electronics Lab	1	none	PHY2006	
PRA2014 Genetics	1	Core 1000 level courses + BIO2001	BIO2007	
PRA2015 Advanced Academic Skills	1	none	none	
PRA2019 Natural Science Illustration	1	none	none	Recommended to previously pass either PRA2015 or PRA2027
PRA2026 R Programming Language	1	none + A laptop computer	none	
PRA2027 Fundamentals of Science Communication	1	PRO1002: Project Period (1000-level)	none	
PRA3030 Introduction in Mycology	1	none	none	Recommended: General biology
PRA3032 Fundamentals of Organic Chemistry Laboratory	1	none	CHE2009	NOTE: This course cannot be followed in case you already followed PRA2002 Chemical Synthesis
PRA3018 Molecular Modeling	1	none	CHE3006	
PRA3021 Topics in Scientific Computing	1	MAT2008 + MAT2006 + MAT2007	none	Recommended: KEN0149
PRA3022 Isom Competition	1	PRO1022	none	
PRA3028 Inorganic Synthesis	1	PRA2034	CHE3010	Recommended: The practical complements perfectly Spectroscopy (CHE2004). NOTE: waivers can be granted if you have followed and passed PRA2032 Fundamentals of Organic Chem Laboratory. No waivers are granted to follow PRA3028 without CHE3010.
PRA3030 Finite Element Method	1	none	none	Recommended: Some experience with Calculus, Linear Algebra, mathematical modeling and/or programming is recommended
PRA3201 Island Studies: Fieldwork and Data Collection	1	PRA2009 + INT1007 + INT2012 + Previously passed + 3000 module in either Biology, Archeology, or Paleontology.	INT3201 + INT3202	
PRA3503 Microbiome Analysis	1	INT2007 or MAT2007 and/or Basic familiarity with computer programming (R) and command line interfaces	none	Recommended: BIO3003 and/or BIO3010 and/or INT3007

Courses #2	Period	Pre-requisites	Co-requisites	Special requirements
BIO1101 Introduction to Biology	2	none	PRA1102	
BIO2009 Great Transformations in Vertebrate Evolution	2	none	none	Recommended: BIO2005
BIO3001 Molecular Biology	2	BIO2001 + BIO2007	PRA3003	
BIO3007 Tropical Biology	2	BIO2002	none	
CHE2006 Biochemistry	2	BIO1101 or CHE1101	none	
CHE2011 Organic Chemistry	2	CHE2009	PRA2033	Recommended: CHE2004. Note: This course cannot be followed in case you already followed CHE2001 Organic Chemistry + CHE3001 Organic Reactions
CHE3007 Chemical Kinetics	2	MAT2006	none	Recommended: PHY2002
CHE3013 Advanced Organic Chemistry	2	CHE2011	PRA3029	
INT1101 Introduction to Liberal Arts and Sciences	2	none	PRA1102	
INT2007 Science in Action	2	none	none	
INT3005 Biobased Materials and Technology	2	CHE2011	none	
INT3007 Systems Biology	2	PRO1002	none	We strongly advise students to have followed at least one level 2000 course in biology, chemistry, mathematics, or neuroscience. Third year OACS students interested in following the course are welcome to contact us about a waiver for the pre-requisite
INT3009 Chemical Ecology **	2	BIO2001 or CHE2009	none	Attending #OCS02. Offered in 26/27
INT3012 Science Education: Model-Based Inquiry	2	INT2013	PRA3026	
MAT1006 Applied Statistics	2	none	none	
MAT2009 Multivariable Calculus	2	MAT2006	none	Recommended: MAT2004 Linear Algebra (highly recommended)
NEU2004 Neural Circuits and Dynamics	2	none	none	
NEU3003 Brain-Body Interactions	2	NEU1001	none	Recommended: NEU2005 + Basic understanding of human biology, e.g., BIO2010
PHY2003 Vibrations and Waves	2	PHY2001	none	
PHY2010 Galactic Astronomy	2	none	none	Recommended: PHY2009
PHY3004 Elementary Particle Physics	2	MAT2004 + PHY2005	none	Recommended: MAT2004 / PHY3001 / PHY3002
PHY3008 Cosmology	2	MAT2009 + PHY2011	none	Recommended: At least one Astronomy course (PHY2008/PHY2009/PHY2010)
PRA1102 Introduction to Scientific Research II	2	MSF1000	BIO1101 + INT1101	
PRA1003 Basic Physics Laboratory	2	none	none	
PRA2010 Synthetic Biology	2	PRA2005 + PRA2014	none	
PRA2013 Practical Zoology	2	BIO2004 + PRA1102	none	Please note that in this course you are required to carry out experimental work with live (non-protected) animals, which may harm them
PRA2018 Capita Selecta in Modern Sciences	2	none	none	
PRA2025 Vertebrate Functional Morphology	2	none	BIO2008	
PRA3033 Organic Chemistry Laboratory	2	PRA2032	CHE2011	Recommended: CHE2004. This course cannot be followed in case you already followed PRA2002 Chemical Synthesis + PRA3001 Advanced Organic Synthesis
PRA3002 Advanced Physics Laboratory	2	PRA1003 + PRA2007	none	Note: Waivers are unlikely to be granted unless applicants show significant expertise in experimental physics or similar subject (equivalent to the two pre-requisite skills)
PRA3003 Molecular Biology	2	PRA2014 + 1000-level core courses	BIO3001	
PRA3005 Polymer Processing	2	CHE2001	none	Note: This skill will require students to arrange their own travel to the Geleen Chemistat campus to attend laboratories
PRA3006 Programming in the Life Sciences	2	MAT2007 OR PRA2003	none	
PRA3023 Plant Physiology and Microbiomes	2	BIO2003 OR a microbiology course	none	
PRA3026 Science Teaching Skills	2	INT2013	INT3012	
PRA3027 Advanced Functional Morphology	2	PRA2025	none	
PRA3029 Advanced Organic Chemistry Laboratory	2	CHE2011	CHE3013	
PRA3031 Statistical Modeling	2	none	none	

* Offered in alternate years (odd years, only in 2027-2028)

** Offered in alternate years (even years, only in 2026-2027)